Chapter 9. The Fundamental Group

Path homotopy, covering spaces, the circle, retractions and fixed points, Borsuk–Ulam, homotopy type, spheres and surfaces

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9.1 Homotopy of paths

Motivation. A loop around the hole of the punctured plane cannot be shrunk; homotopy makes this precise and turns spaces into groups.

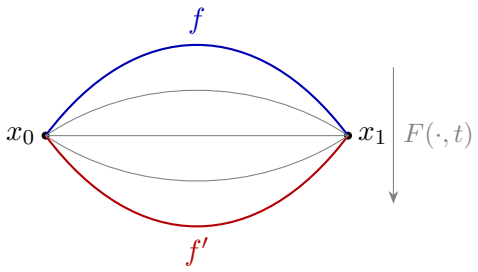


Figure. *A path homotopy F from f to f' with end points x_0, x_1 fixed.*

Maps $f, f' : X \to Y$ are **homotopic**, $f \simeq f'$, if there is a continuous $F : X \times I \to Y$ ($I = [0,1]$) with $F(x,0) = f(x)$, $F(x,1) = f'(x)$; f is **nulhomotopic** if it is homotopic to a constant. Paths $f, f' : I \to X$ from x_0 to x_1 are **path homotopic**, $f \simeq_p f'$, if there is $F : I \times I \to X$ with $F(s,0) = f(s)$, $F(s,1) = f'(s)$, $F(0,t) = x_0$, $F(1,t) = x_1$.

Lemma 9.1 (51.1)

$\simeq$ and $\simeq_p$ are equivalence relations.

Proof. $F(x,t) = f(x)$ gives $f \simeq f$; if $F : f \simeq f'$, then $F(x, 1-t)$ gives $f' \simeq f$. For $F : f \simeq f'$, $F' : f' \simeq f''$, let $G(x,t) = F(x,2t)$ $(t \le \frac{1}{2})$, $F'(x, 2t-1)$ $(t \ge \frac{1}{2})$, continuous by the pasting lemma. For paths these keep the end points fixed. ∎

$[f]$ denotes the path-homotopy class of f.

Example 9.1

For maps (or paths with the same end points) f, g into a convex subset of $\mathbb{R}^n$, the **straight-line homotopy** $F(x,t) = (1-t)f(x) + tg(x)$ is a homotopy (path homotopy). In $\mathbb{R}^2 - \{0\}$, however, $f(s) = (\cos \pi s, \sin \pi s)$ and $g(s) = (\cos \pi s, -\sin \pi s)$ are not path homotopic (Section 9.4).

For a path f from x_0 to x_1 and g from x_1 to x_2, the **product** $f * g$ is $f(2s)$ on $[0, \frac{1}{2}]$ and $g(2s-1)$ on $[\frac{1}{2}, 1]$. If $F : f \simeq_p f'$ and $G : g \simeq_p g'$, then $H(s,t) = F(2s,t)$ $(s \le \frac{1}{2})$, $G(2s-1,t)$ $(s \ge \frac{1}{2})$ is a path homotopy $f * g \simeq_p f' * g'$, so $[f] * [g] = [f * g]$ is defined. For continuous $k : X \to Y$, $k \circ F$ is a path homotopy from $k \circ f$ to $k \circ f'$, and $k \circ (f * g) = (k \circ f) * (k \circ g)$.

Theorem 9.2 (51.2)

The operation $*$ on path-homotopy classes satisfies:
(1) (associativity) if $[f] * ([g] * [h])$ is defined, so is $([f] * [g]) * [h]$, and they are equal;
(2) (identities) if e_x is the constant path at x and f runs from x_0 to x_1, then $[e_{x_0}] * [f] = [f] = [f] * [e_{x_1}]$;
(3) (inverses) with $\bar{f}(s) = f(1-s)$, $[f] * [\bar{f}] = [e_{x_0}]$ and $[\bar{f}] * [f] = [e_{x_1}]$.

Proof. Let e_0 be the constant path in I at 0, $i : I \to I$ the identity path, $\bar{i}$ its reverse.

(2) $e_0 * i \simeq_p i$ in the convex I by some G; $f \circ G$ is a path homotopy from $f \circ (e_0 * i) = e_{x_0} * f$ to f. The other identity uses $i * e_1$.

(3) $i * \bar{i} \simeq_p e_0$ in I by some H; $f \circ H$ gives $f * \bar{f} \simeq_p e_{x_0}$. Similarly $\bar{f} * f \simeq_p e_{x_1}$.

(1) For $0 < a < b < 1$ let $k_{a,b}$ be, on $[0,a]$, $[a,b]$, $[b,1]$, the positive linear map onto I followed by f, g, h. Then $(f * g) * h = k_{1/2,3/4}$, $f * (g * h) = k_{1/4,1/2}$. For the piecewise linear $\gamma : I \to I$ with $0, a, b, 1 \mapsto 0, c, d, 1$, $k_{c,d} \circ \gamma = k_{a,b}$; as $\gamma \simeq_p i$ in I, $k_{a,b} \simeq_p k_{c,d}$. ■

Theorem 9.3 (51.3)

Let f be a path in X and $0 = a_0 < a_1 < \cdots < a_n = 1$. Let f_i be the positive linear map of I onto $[a_{i-1}, a_i]$ followed by f. Then $[f] = [f_1] * \cdots * [f_n]$.

Proof. A bracketing of $f_1 * \cdots * f_n$ is, on the intervals of some subdivision $0 = b_0 < \cdots < b_n = 1$, the linear map onto I followed by f_i; so it equals $f \circ \gamma$, γ the piecewise linear map of $[b_{i-1}, b_i]$ onto $[a_{i-1}, a_i]$. As $\gamma \simeq_p i$, $f \circ \gamma \simeq_p f$. ■

9.2 The fundamental group

Motivation. Loop classes at a base point form a group, a homeomorphism invariant.

A **loop** at x_0 is a path from x_0 to x_0. By Theorem 9.2 the classes of loops at x_0 form a group under $*$, the **fundamental group** $\pi_1(X, x_0)$; it is trivial for convex $X \subset \mathbb{R}^n$. For a path α from x_0 to x_1 let $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$, $\hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha]$.

Theorem 9.4 (52.1)

$\hat{\alpha}$ is a group isomorphism.

Proof. $\hat{\alpha}([f]) * \hat{\alpha}([g]) = [\bar{\alpha}] * [f] * [\alpha] * [\bar{\alpha}] * [g] * [\alpha] = \hat{\alpha}([f] * [g])$. With $\beta = \bar{\alpha}$, $\hat{\beta}(\hat{\alpha}([h])) = [\alpha] * [\bar{\alpha}] * [h] * [\alpha] * [\bar{\alpha}] = [h]$; similarly $\hat{\alpha} \circ \hat{\beta} = \mathrm{id}$. ■

Corollary 9.5 (52.2)

If X is path connected and $x_0, x_1 \in X$, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.

For the path component C of x_0, $\pi_1(C, x_0) = \pi_1(X, x_0)$. A path-connected space with trivial fundamental group is **simply connected**.

Lemma 9.6 (52.3)

In a simply connected space, any two paths with the same initial and final points are path homotopic.

Proof. For f, g from x_0 to x_1, $[f] * [\bar{g}] = [e_{x_0}]$; multiply on the right by $[g]$. ■

A continuous $h : (X, x_0) \to (Y, y_0)$ (i.e. $h(x_0) = y_0$) induces the homomorphism $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $h_*([f]) = [h \circ f]$.

Theorem 9.7 (52.4)

If $h : (X, x_0) \to (Y, y_0)$ and $k : (Y, y_0) \to (Z, z_0)$ are continuous, then $(k \circ h)_* = k_* \circ h_*$. If i is the identity of (X, x_0), then i_* is the identity.

Proof. $(k \circ h)_*([f]) = [k \circ (h \circ f)] = k_*(h_*([f]))$, and $i_*([f]) = [f]$. ■

Corollary 9.8 (52.5)

If $h : (X, x_0) \to (Y, y_0)$ is a homeomorphism, h_* is an isomorphism.

Proof. With $k = h^{-1}$, $k_* h_* = (k \circ h)_* = \mathrm{id}$ and $h_* k_* = \mathrm{id}$. ■

9.3 Covering spaces

Motivation. A covering unwraps loops into paths whose end points detect winding, modelled on $\mathbb{R} \to S^1$.

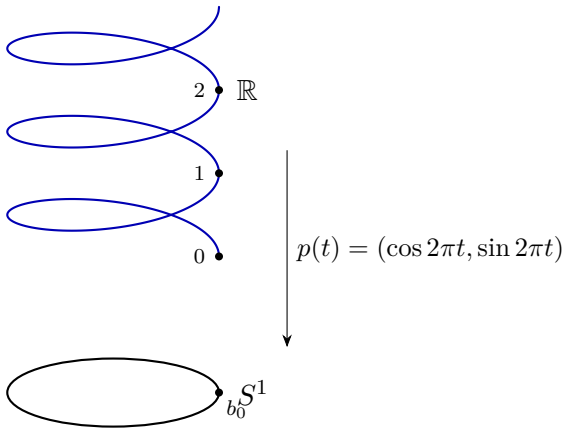


Figure. *$p : \mathbb{R} \to S^1$ with fibre $p^{-1}(b_0) = \mathbb{Z}$; a loop at b_0 lifts to a path from 0 to its class in $\pi_1(S^1) \cong \mathbb{Z}$.*

Let $p : E \to B$ be continuous and surjective. An open $U \subset B$ is **evenly covered** if $p^{-1}(U)$ is a union of disjoint open V_α (**slices**), each mapped homeomorphically onto U. If every point of B has an evenly covered neighbourhood, p is a **covering map** and E a **covering space** of B. A covering map is a local homeomorphism and open, and each fibre $p^{-1}(b)$ is discrete.

Theorem 9.9 (53.1)

$p : \mathbb{R} \to S^1$, $p(x) = (\cos 2\pi x, \sin 2\pi x)$, is a covering map.

Proof. For the open arc U with positive first coordinate, $p^{-1}(U) = \bigcup_n (n - \frac{1}{4}, n + \frac{1}{4})$. p is injective on $[n - \frac{1}{4}, n + \frac{1}{4}]$ (second coordinate increasing), so maps it homeomorphically onto the closed right half circle and $(n - \frac{1}{4}, n + \frac{1}{4})$ onto U. Likewise for the arcs with negative first, positive second and negative second coordinate, which with U cover S^1. ■

Example 9.2

$p|(0, \infty)$ is a surjective local homeomorphism but not a covering map: for any neighbourhood U of $b_0 = (1, 0)$, the component $(0, \epsilon)$ of $p^{-1}(U)$ is not mapped onto U.

Theorem 9.10 (53.2)

If $p : E \to B$ is a covering map, $B_0 \subset B$ and $E_0 = p^{-1}(B_0)$, then $p_0 = p|E_0 : E_0 \to B_0$ is a covering map.

Proof. For $b \in B_0$ and evenly covered $U \ni b$ with slices V_α, $p_0^{-1}(U \cap B_0)$ is the disjoint union of the open $V_\alpha \cap E_0$, each mapped homeomorphically onto $U \cap B_0$. ■

Theorem 9.11 (53.3)

If $p : E \to B$ and $p' : E' \to B'$ are covering maps, so is $p \times p' : E \times E' \to B \times B'$.

Proof. For evenly covered $U \ni b$, $U' \ni b'$ with slices V_α, V'_β, $(p \times p')^{-1}(U \times U')$ is the disjoint union of the open $V_\alpha \times V'_\beta$, each mapped homeomorphically onto $U \times U'$. ■

Example 9.3

$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$ covers the torus. Restricting to $b_0 \times S^1 \cup S^1 \times b_0$ (Theorem 9.10) gives a covering of the figure eight by the grid of horizontal and vertical lines through integer points. As $\mathbb{R}_+ \times S^1 \cong \mathbb{R}^2 - 0$ (polar coordinates), $\mathrm{id} \times p : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^2 - 0$ covers the punctured plane.

9.4 The fundamental group of the circle

Motivation. The covering $\mathbb{R} \to S^1$ gives $\pi_1(S^1) \cong \mathbb{Z}$, on which the rest of the chapter builds.

For $p : E \to B$, a **lifting** of $f : X \to B$ is a map $\tilde{f} : X \to E$ with $p \circ \tilde{f} = f$.

Lemma 9.12 (54.1)

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Any path $f : I \to B$ beginning at b_0 has a unique lifting $\tilde{f}$ beginning at e_0.

Proof. By the Lebesgue number lemma choose $0 = s_0 < \cdots < s_n = 1$ with each $f([s_i, s_{i+1}])$ in an evenly covered U.

Existence: $\tilde{f}(0) = e_0$; if $\tilde{f}$ is defined on $[0, s_i]$, let V_0 be the slice containing $\tilde{f}(s_i)$ and $\tilde{f}(s) = (p|V_0)^{-1}(f(s))$ on $[s_i, s_{i+1}]$ (pasting lemma).

Uniqueness: if another lifting $\tilde{f}'$ equals $\tilde{f}$ on $[0, s_i]$, the connected $\tilde{f}'([s_i, s_{i+1}])$ lies in one slice, V_0 as it contains $\tilde{f}(s_i)$; so $\tilde{f}' = (p|V_0)^{-1} \circ f = \tilde{f}$ there. ■

Lemma 9.13 (54.2)

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$, and $F : I \times I \to B$ continuous with $F(0,0) = b_0$. There is a unique lifting $\tilde{F} : I \times I \to E$ with $\tilde{F}(0,0) = e_0$. If F is a path homotopy, so is $\tilde{F}$.

Proof. Existence: choose subdivisions $s_0 < \cdots < s_m$, $t_0 < \cdots < t_n$ with each $I_i \times J_j = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ mapped into an evenly covered set (Lebesgue). Lift on $A = 0 \times I \cup I \times 0$ by Lemma 9.12, then over the rectangles in the order $I_1 \times J_1, \ldots, I_m \times J_1, I_1 \times J_2, \ldots$. For the next rectangle R, $F(R) \subset U$ evenly covered, and $C = R \cap A'$ ($A' = A$ plus the rectangles done) is its connected left and bottom edge; $\tilde{F}(C)$ lies in one slice V_0, and $\tilde{F} = (p|V_0)^{-1} \circ F$ on R agrees on C and is continuous on $A' \cup R$ (pasting lemma). Uniqueness: a lifting is determined on A (Lemma 9.12), and on each R its connected image meets V_0, so lies in V_0 and equals $(p|V_0)^{-1} \circ F$. If F is a path homotopy, $\tilde{F}(0 \times I)$ is a connected subset of the discrete fibre $p^{-1}(b_0)$, a point; likewise for $1 \times I$. ■

Theorem 9.14 (54.3)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and f, g paths in B from b_0 to b_1 with liftings $\tilde{f}, \tilde{g}$ beginning at e_0. If $f \simeq_p g$, then $\tilde{f}$ and $\tilde{g}$ end at the same point and $\tilde{f} \simeq_p \tilde{g}$.

Proof. Lift $F : f \simeq_p g$ to $\tilde{F}$ from e_0, a path homotopy (Lemma 9.13). $\tilde{F}|I \times 0$ lifts f from e_0, so is $\tilde{f}$; $\tilde{F}|I \times 1$ lifts g from $\tilde{F}(0,1) = e_0$, so is $\tilde{g}$. As $\tilde{F}(1 \times I)$ is a point, $\tilde{f}(1) = \tilde{g}(1)$. ■

The **lifting correspondence** $\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$, $\phi([f]) = \tilde{f}(1)$ with $\tilde{f}$ the lifting from e_0, is well defined by Theorem 9.14.

Theorem 9.15 (54.4)

If E is path connected, ϕ is surjective; if E is simply connected, ϕ is bijective.

Proof. For $e_1 \in p^{-1}(b_0)$ and a path γ in E from e_0 to e_1, γ lifts $f = p \circ \gamma$, so $\phi([f]) = e_1$. If $\phi([f]) = \phi([g])$ and E is simply connected, $\tilde{f} \simeq_p \tilde{g}$ by some $\tilde{F}$ (Lemma 9.6), and $p \circ \tilde{F} : f \simeq_p g$. ■

Theorem 9.16 (54.5)

$\pi_1(S^1, b_0)$ is isomorphic to the additive group $\mathbb{Z}$.

Proof. For $p : \mathbb{R} \to S^1$, $e_0 = 0$: $p^{-1}(b_0) = \mathbb{Z}$ and $\mathbb{R}$ is simply connected, so ϕ is a bijection onto $\mathbb{Z}$. If $\tilde{f}, \tilde{g}$ lift f, g from 0 with $\tilde{f}(1) = n$, $\tilde{g}(1) = m$, then $n + \tilde{g}$ lifts g from n ($p(n + x) = p(x)$), so $\tilde{f} * (n + \tilde{g})$ lifts $f * g$ from 0 to $n + m$: $\phi([f] * [g]) = \phi([f]) + \phi([g])$. ■

So $p_0(s) = p(s)$ generates $\pi_1(S^1, b_0)$, and a loop's class is the end point n of its lifting, the number of times it winds around.

Theorem 9.17 (54.6)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$.
(a) $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective.
(b) Let $H = p_*(\pi_1(E, e_0))$. The lifting correspondence induces an injective map $\Phi : \pi_1(B, b_0)/H \to p^{-1}(b_0)$ of the right cosets of H, bijective if E is path connected.
(c) A loop f at b_0 has $[f] \in H$ iff f lifts to a loop at e_0.

Proof. (a) If $\tilde{h}$ is a loop at e_0 and $F : p \circ \tilde{h} \simeq_p e_{b_0}$, the lifting $\tilde{F}$ from e_0 is a path homotopy between $\tilde{h}$ and the constant loop.

(c) If $[f] = [p \circ \tilde{h}]$, $\tilde{h}$ a loop at e_0, then $\tilde{f}$ ends where $\tilde{h}$ does, at e_0 (Theorem 9.14). Conversely, if $\tilde{f}$ is a loop, $[f] = p_*[\tilde{f}]$.

(b) Let $\Phi(H * [f]) = \phi([f])$. For $[h] \in H$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at e_0, $\tilde{h} * \tilde{f}$ lifts $h * f$ and ends at $\tilde{f}(1)$; so Φ is well defined. If $\phi([f]) = \phi([g])$, $\tilde{f} * \bar{\tilde{g}}$ is a loop at e_0 lifting $f * \bar{g}$, so $[f] * [g]^{-1} \in H$, $H * [f] = H * [g]$. Surjectivity: Theorem 9.15. ■

9.5 Retractions and fixed points

Motivation. $\pi_1(S^1) \neq 0$ forbids retracting the disc onto its boundary, giving Brouwer's theorem in dimension two.

$A \subset X$ is a **retract** of X if there is a continuous $r : X \to A$ with $r|A = \mathrm{id}_A$ (a **retraction**).

Lemma 9.18 (55.1)

If A is a retract of X, the homomorphism j_* induced by inclusion $j : A \to X$ is injective.

Proof. $r \circ j = \mathrm{id}_A$, so $r_* \circ j_* = \mathrm{id}$. ■

Theorem 9.19 (55.2, no-retraction theorem)

There is no retraction of B^2 onto S^1.

Proof. $j_* : \pi_1(S^1) \cong \mathbb{Z} \to \pi_1(B^2) = 0$ is not injective. ■

Lemma 9.20 (55.3)

For continuous $h : S^1 \to X$ the following are equivalent:
(1) h is nulhomotopic;
(2) h extends to a continuous $k : B^2 \to X$;
(3) h_* is the trivial homomorphism.

Proof. (1)⇒(2): let $H : S^1 \times I \to X$ be a homotopy from h to a constant. $\pi(x,t) = (1-t)x$, $S^1 \times I \to B^2$, is continuous, closed and onto, a quotient map collapsing only $S^1 \times 1$ (to 0). H is constant on $S^1 \times 1$, so induces k with $k \circ \pi = H$, extending h. (2)⇒(3): $h = k \circ j$, $j : S^1 \to B^2$, and j_* is trivial. (3)⇒(1): for the standard loop p_0, $b_0 = p_0(0)$, $h \circ p_0 \simeq_p$ const by some F, with $F(0,t) = F(1,t) = F(s,1) = h(b_0)$. The quotient map $q(s,t) = (1-t)p_0(s)$, $I \times I \to B^2$, identifies $(0,t) \sim (1,t)$ and collapses $I \times 1$; F respects this, inducing $k : B^2 \to X$ extending h. Then $H(x,t) = k((1-t)x)$ is a homotopy from h to the constant $k(0)$. ■

Corollary 9.21 (55.4)

The inclusion $j : S^1 \to \mathbb{R}^2 - 0$ and the identity $i : S^1 \to S^1$ are not nulhomotopic.

Proof. $x \mapsto x/\|x\|$ retracts $\mathbb{R}^2 - 0$ onto S^1, so j_* is injective (Lemma 9.18), hence nontrivial; i_* is the identity of $\mathbb{Z}$. Apply Lemma 9.20. ■

Theorem 9.22 (55.5)

Given a nonvanishing vector field on B^2 (a continuous $v : B^2 \to \mathbb{R}^2 - 0$), there is a point of S^1 where v points directly inward and one where it points directly outward.

Proof. Suppose $v(x) \neq ax$ for all $x \in S^1$, $a < 0$. $w = v|S^1 : S^1 \to \mathbb{R}^2 - 0$ extends to B^2, so is nulhomotopic. $F(x,t) = tx + (1-t)w(x)$ never vanishes on $S^1 \times I$ (for $0 < t < 1$, $F = 0$ means $w(x) = -\frac{t}{1-t}x$), so $j \simeq w$ is nulhomotopic, contradicting Corollary 9.21. For outward, use $-v$. ■

Theorem 9.23 (55.6, Brouwer fixed-point theorem for the disc)

Every continuous $f : B^2 \to B^2$ has a fixed point.

Proof. Otherwise $v(x) = f(x) - x$ is a nonvanishing vector field. At $x \in S^1$, $f(x) - x = ax$ with $a > 0$ would give $f(x) = (1+a)x \notin B^2$, so v never points directly outward, contradicting Theorem 9.22. ■

Corollary 9.24 (55.7)

A 3×3 matrix A of positive real numbers has a positive real eigenvalue.

Proof. $B = \{x \in S^2 : x_1, x_2, x_3 \geq 0\}$ is homeomorphic (by $x \mapsto x/(x_1 + x_2 + x_3)$) to a closed triangle, hence to B^2. Ax has positive entries, so $x \mapsto Ax/\|Ax\|$ maps B into B; a fixed point gives $Ax = \|Ax\|x$. ■

Theorem 9.25 (55.8)

Let T be a closed triangle in $\mathbb{R}^2$. There is $\epsilon > 0$ such that for every open covering $\mathcal{A}$ of T by sets of diameter less than ϵ, some point of T belongs to at least three elements of $\mathcal{A}$.

Proof. No point lies on all three edges e_1, e_2, e_3, so $(a, b, c) \mapsto \max(|a-b|, |b-c|, |a-c|)$ has a positive minimum ϵ on the compact $e_1 \times e_2 \times e_3$, and no set of diameter $< \epsilon$ meets all three edges. Suppose no point lies in three elements of such an $\mathcal{A}$; pass to a finite subcovering with a dominated partition of unity $\{\phi_A\}$, and choose a vertex v_A with A missing the opposite edge. Let $k(x) = \sum_A \phi_A(x) v_A$: a convex combination of at most two vertices, so in ∂T. If $x \in e$ and $\phi_A(x) > 0$, A meets e, so $v_A \in e$; thus $k(e) \subset e$, and $H(x, t) = (1 - t)x + tk(x)$ is a homotopy in ∂T from $\mathrm{id}_{\partial T}$ to $k|\partial T$. But $k|\partial T$ extends to T, so is nulhomotopic (Lemma 9.20, $T \cong B^2$), and then so is $\mathrm{id}_{\partial T}$, $\partial T \cong S^1$, contradicting Corollary 9.21. ■

So the triangle has topological dimension at least 2.

9.6 The fundamental theorem of algebra

Motivation. A degree-n polynomial winds a large circle n times around 0, impossible if it had no root.

Theorem 9.26 (56.1, fundamental theorem of algebra)

A polynomial equation $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ of degree $n > 0$ with real or complex coefficients has a complex root.

Proof.

Step 1. $f(z) = z^n$, $S^1 \to S^1$, induces an injective homomorphism: $f \circ p_0(s) = p(ns)$ lifts to $s \mapsto ns$, so f_* is multiplication by n.

Step 2. $g(z) = z^n$, $S^1 \to \mathbb{R}^2 - 0$, equals $j \circ f$, so $g_* = j_* f_*$ is injective and g is not nulhomotopic.

Step 3. If $\sum |a_i| < 1$, there is a root in B^2. Otherwise $k(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ maps B^2 into $\mathbb{R}^2 - 0$, so $h = k|S^1$ is nulhomotopic. But $F(z, t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$ never vanishes on $S^1 \times I$, as $|t(a_{n-1}z^{n-1} + \cdots + a_0)| \leq \sum |a_i| < 1 = |z^n|$; so $g \simeq h$ in $\mathbb{R}^2 - 0$, a contradiction.

Step 4. In general choose $c > 0$ with $\sum_i |a_{n-i}|/c^i < 1$; $y^n + (a_{n-1}/c)y^{n-1} + \cdots + a_0/c^n = 0$ has a root y_0, and cy_0 is a root of the original. ■

9.7 The Borsuk–Ulam theorem

Motivation. Some point on the earth has the same temperature and pressure as its antipode.

$h : S^n \to S^m$ is **antipode-preserving** if $h(-x) = -h(x)$.

Theorem 9.27 (57.1)

A continuous antipode-preserving $h : S^1 \to S^1$ is not nulhomotopic.

Proof. After a rotation (commuting with $x \mapsto -x$), $h(b_0) = b_0$. In $S^1 \subset \mathbb{C}$, $q(z) = z^2$ is a quotient map identifying z with $-z$; as $q(h(-z)) = q(h(z))$, there is a continuous k with $k \circ q = q \circ h$. q_* is injective ($q \circ p_0$ lifts to $s \mapsto 2s$). For $f(s) = e^{\pi i s}$, from b_0 to $-b_0$, $q \circ f = p_0$; $h \circ f$ runs from b_0 to $-b_0$, so its lifting $\tilde{g}$ from 0 ends at some $m + \frac{1}{2}$, and $q \circ h \circ f = p \circ 2\tilde{g}$ with $2\tilde{g}$ ending at $2m+1$. So $k_*[p_0] = [q \circ h \circ f]$ corresponds to $2m+1 \neq 0$ and k_* is injective. If h were nulhomotopic, $k_* q_* = q_* h_*$ would be trivial, a contradiction. ■

Theorem 9.28 (57.2)

There is no continuous antipode-preserving map $g : S^2 \to S^1$.

Proof. Its restriction h to the equator is antipode-preserving, hence not nulhomotopic, but extends to the upper hemisphere $\cong B^2$ (via $(x, y, z) \mapsto (x, y)$), so is nulhomotopic (Lemma 9.20). ■

Theorem 9.29 (57.3, Borsuk–Ulam theorem for S^2)

For every continuous $f : S^2 \to \mathbb{R}^2$ there is $x \in S^2$ with $f(x) = f(-x)$.

Proof. Otherwise $g(x) = (f(x) - f(-x))/\|f(x) - f(-x)\|$ is a continuous antipode-preserving map $S^2 \to S^1$. ■

Theorem 9.30 (57.4, bisection theorem)

Given two bounded polygonal regions in $\mathbb{R}^2$, some line bisects each of them.

Proof. Put A_1, A_2 in the plane $z = 1$ of $\mathbb{R}^3$. For $u \in S^2$ let $f_i(u)$ be the area of the part of A_i with $x \cdot u > 0$; $f_i(-u) = \text{area}(A_i) - f_i(u)$, and f_i is continuous (assumed). Borsuk–Ulam for (f_1, f_2) gives u with $f_i(u) = \frac{1}{2}\,\text{area}(A_i)$. As $u \neq \pm(0, 0, 1)$ (there $f_i(u)$ is 0 or the full area), the plane $x \cdot u = 0$ meets $z = 1$ in a line bisecting both regions. ■

9.8 Deformation retracts and homotopy type

Motivation. Homotopy equivalence lets us replace a space by a simpler one, e.g. $\mathbb{R}^2 - 0$ by S^1.

Lemma 9.31 (58.1)

If $h, k : (X, x_0) \to (Y, y_0)$ are homotopic by a homotopy keeping x_0 at y_0, then $h_* = k_*$.

Proof. For a loop f at x_0, $G(s, t) = H(f(s), t)$ is a path homotopy from $h \circ f$ to $k \circ f$. ■

Theorem 9.32 (58.2)

The inclusion $j : S^n \to \mathbb{R}^{n+1} - 0$ induces an isomorphism of fundamental groups.

Proof. $r(x) = x/\|x\|$ has $r \circ j = \text{id}$, and $H(x, t) = (1 - t)x + tx/\|x\|$ is a homotopy in $\mathbb{R}^{n+1} - 0$ from the identity to $j \circ r$ fixing S^n. By Lemma 9.31, $j_* r_* = \text{id}$, and $r_* j_* = \text{id}$. ■

$A \subset X$ is a **deformation retract** of X if there is $H : X \times I \to X$ with $H(x,0) = x$, $H(x,1) \in A$, $H(a,t) = a$ for $a \in A$.

Theorem 9.33 (58.3)

If A is a deformation retract of X and $x_0 \in A$, then $j : (A, x_0) \to (X, x_0)$ induces an isomorphism.

Proof. As in Theorem 9.32, with $r(x) = H(x,1)$. ■

Example 9.4

$\mathbb{R}^2 - \{p, q\}$ deformation retracts onto a figure eight and onto a theta space; the solid torus $B^2 \times S^1$ and $\mathbb{R}^3$ minus the z-axis deformation retract onto a circle, so have $\pi_1 \cong \mathbb{Z}$.

$f : X \to Y$ is a **homotopy equivalence** with **homotopy inverse** $g : Y \to X$ if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$; then X, Y have the same **homotopy type**.

Lemma 9.34 (58.4)

Let $h, k : X \to Y$ be homotopic by H, with $h(x_0) = y_0$, $k(x_0) = y_1$, and let $\alpha(t) = H(x_0, t)$. Then $k_* = \hat{\alpha} \circ h_*$.

Proof. For a loop f at x_0 let $G(s,t) = H(f(s), t)$. In the convex $I \times I$, $\beta_0 * \gamma_1 \simeq_p \gamma_0 * \beta_1$, where $\beta_j(s) = (s, j)$, $\gamma_j(t) = (j, t)$; applying G, $(h \circ f) * \alpha \simeq_p \alpha * (k \circ f)$, so $[k \circ f] = [\bar{\alpha}] * [h \circ f] * [\alpha]$. ■

Corollary 9.35 (58.5)

If $h \simeq k$ and h_* is injective, surjective or trivial, so is k_*.

Corollary 9.36 (58.6)

If h is nulhomotopic, h_* is trivial.

Proof. Both follow from Lemma 9.34: $\hat{\alpha}$ is an isomorphism, and constant maps induce the trivial homomorphism. ■

Theorem 9.37 (58.7)

If $f : X \to Y$ is a homotopy equivalence with $f(x_0) = y_0$, then $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ is an isomorphism.

Proof. Let g be a homotopy inverse, $x_1 = g(y_0)$, $y_1 = f(x_1)$. By Lemma 9.34, $g \circ f \simeq \mathrm{id}_X$ makes $g_* \circ (f_{x_0})_* : \pi_1(X, x_0) \to \pi_1(X, x_1)$ an isomorphism, so $(g_{y_0})_*$ is onto; $f \circ g \simeq \mathrm{id}_Y$ makes $(f_{x_1})_* \circ (g_{y_0})_*$ an isomorphism, so $(g_{y_0})_*$ is injective. Hence $(g_{y_0})_*$, and so $(f_{x_0})_*$, is an isomorphism. ■

9.9 The fundamental group of S^n

Motivation. Higher spheres are simply connected, by computing π_1 from an open cover (made systematic in Chapter 11).

Theorem 9.38 (59.1)

Let $X = U \cup V$ with U, V open, $U \cap V$ path connected and $x_0 \in U \cap V$. Then the images of $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ and $j_* : \pi_1(V, x_0) \to \pi_1(X, x_0)$ generate $\pi_1(X, x_0)$.

Proof. For a loop f at x_0 choose (Lebesgue) $0 = a_0 < \cdots < a_n = 1$ with each $f([a_{i-1}, a_i])$ in U or V. If some $f(a_i)$, $0 < i < n$, lies in $U - V$, say, both adjacent intervals map into U and a_i can be deleted; so assume all $f(a_i) \in U \cap V$. Choose paths α_i in $U \cap V$ from x_0 to $f(a_i)$ (α_0, α_n constant). With f_i as in Theorem 9.3, $[f] = [f_1] * \cdots * [f_n] = [g_1] * \cdots * [g_n]$, where $g_i = \alpha_{i-1} * f_i * \bar{\alpha}_i$ is a loop at x_0 in U or V. ∎

Corollary 9.39 (59.2)

If $X = U \cup V$ with U, V open and simply connected and $U \cap V$ nonempty and path connected, then X is simply connected.

Theorem 9.40 (59.3)

For $n \geq 2$, S^n is simply connected.

Proof. Let $p = (0, \ldots, 0, 1)$. Stereographic projection $f(x) = (x_1, \ldots, x_n)/(1 - x_{n+1})$ is a homeomorphism $S^n - p \to \mathbb{R}^n$ with inverse $y \mapsto (t(y)y, 1 - t(y))$, $t(y) = 2/(1 + \|y\|^2)$; similarly $S^n - (-p) \cong \mathbb{R}^n$. Both are simply connected, and their intersection is $\cong \mathbb{R}^n - 0$, path connected for $n \geq 2$. Apply Corollary 9.39. ∎

9.10 Fundamental groups of some surfaces

Motivation. π_1 distinguishes the sphere, torus, projective plane and double torus.

A **surface** is a Hausdorff space with a countable basis in which each point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^2$.

Theorem 9.41 (60.1)

$\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$.

Proof. With projections p, q, $\Phi([f]) = (p_*[f], q_*[f])$ is a homomorphism. For loops g in X, h in Y, $f(s) = g(s) \times h(s)$ maps to $([g], [h])$. If $G : p \circ f \simeq_p e$ and $H : q \circ f \simeq_p e$, then $G \times H : f \simeq_p e$. ∎

Corollary 9.42 (60.2)

The fundamental group of the torus $S^1 \times S^1$ is $\mathbb{Z} \times \mathbb{Z}$.

The **projective plane** P^2 is the quotient of S^2 obtained by identifying each x with $-x$.

Theorem 9.43 (60.3)

P^2 is a compact surface, and the quotient map $p : S^2 \to P^2$ is a covering map.

Proof. p is open: $p^{-1}(p(U)) = U \cup a(U)$, $a(x) = -x$. For $x \in S^2$ the ϵ-neighbourhood U ($\epsilon < 1$) contains no antipodal pair, so $p|U$ and $p|a(U)$ are continuous open bijections onto $p(U)$, homeomorphisms, and $a(U) \cap U = \emptyset$: $p(U)$ is evenly covered. Hausdorff: for $p(x) \neq p(y)$, $\pm x, \pm y$ are

distinct; for ϵ less than half their minimum distance, the ϵ-neighbourhoods $U \ni x$, $W \ni y$ have $(U \cup a(U)) \cap (W \cup a(W)) = \emptyset$, so $p(U) \cap p(W) = \emptyset$. p open carries a countable basis of S^2 to one of P^2; P^2 is locally homeomorphic to S^2, hence to $\mathbb{R}^2$, and compact as an image of S^2. ■

Corollary 9.44 (60.4)

$\pi_1(P^2, y)$ has order 2.

Proof. S^2 is simply connected, so the lifting correspondence is a bijection onto the two-point fibre (Theorem 9.15). ■

Similarly $\pi_1(P^n)$ $(n \geq 2)$ has order 2.

Lemma 9.45 (60.5)

The fundamental group of the figure eight is not abelian.

Proof. Let $X = A \cup B$, circles meeting at x_0. Let $E \subset \mathbb{R}^2$ be the union of the coordinate axes and, for each integer $n \neq 0$, a circle tangent to the x-axis at $(n, 0)$ and one tangent to the y-axis at $(0, n)$. Let $p : E \to X$ wrap the x-axis around A (each $[n, n+1]$ once, integers to x_0) and the y-axis around B, and map each circle tangent to the x-axis onto B, each tangent to the y-axis onto A (tangency points to x_0). Small arcs in $X - x_0$ and a small "cross" at x_0 have preimages that are disjoint arcs, resp. crosses about the lattice points on the axes, mapped homeomorphically; so p is a covering map. For loops g, h once around A, B, the lifting of $g * h$ from the origin runs to $(1, 0)$ and around the circle tangent there, ending at $(1, 0)$; that of $h * g$ ends at $(0, 1)$. By Theorem 9.14, $g * h \not\simeq_p h * g$. ■

Theorem 9.46 (60.6)

The fundamental group of the double torus is not abelian.

Proof. The double torus X contains a figure eight A (a circle around each handle, meeting at a point), a retract of X: squash each handle onto its circle, compatibly along the joining curve. By Lemma 9.18, $\pi_1(X)$ contains the nonabelian $\pi_1(A)$. ■

Corollary 9.47 (60.7)

The 2-sphere, torus, projective plane and double torus are topologically distinct.

Proof. Their fundamental groups are trivial, $\mathbb{Z} \times \mathbb{Z}$ (infinite abelian), of order 2, and nonabelian. ■